

HARIN DULNETH

SOFTWARE ENGINEER INTERN

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SUMMARY

Computer Science undergraduate specializing in Artificial Intelligence, with a focus on building AI-driven, production-grade systems. Experienced in developing scalable backend services with FastAPI and .NET, and integrating machine learning solutions using PyTorch and TensorFlow. Skilled in developing end-to-end ML pipelines, RESTful APIs and data-driven applications with PostgreSQL, MongoDB and Redis. Comfortable with Docker and CI/CD workflows, ensuring reliable and maintainable deployments. Passionate about leveraging AI to build practical, user-centric and high-performance software solutions.

EDUCATION

Bsc. in Computer Science (Hons)

2023 - Present

University of Kelaniya, Sri Lanka

Specializing in Artificial Intelligence (Current GPA - 3.82)

G.C.E Advanced Level - 1A 2Bs

2021 - 2022

Physical Science Stream

Isipathana College, Colombo 05

G.C.E Ordinary Level - 8As 1B

2017 - 2018

Isipathana College, Colombo 05

CORE COMPETENCIES

- **Programming Languages** - Python | C | C# | Java
JavaScript | TypeScript
- **Backend** - FastAPI | Flask | Node.js | .NET
Spring Boot | Express.js
- **Databases** - SQL | PostgreSQL | Redis
MongoDB | Supabase
- **Machine Learning / AI** - TensorFlow | PyTorch
LangChain | LangGraph | Hugging Face | n8n
- **Frontend** - React | Next.js | Tailwind
- **Other** - Git | Docker | CI/CD | Linux | AWS | Google
Colab | Figma

PROJECTS

Trowd - Real-Time Transit Intelligence Platform | [Project](#)

Technical Stack - React (TypeScript), FastAPI, Node.js, PostgreSQL, Redis, Socket.IO, LightGBM, XGBoost

- Built a full-stack bus tracking system with ML-based ETA forecasting via a fusion engine using confidence-weighted blending of real-time telemetry and historical segment data.
- Engineered a crowd-sourced arrival confirmation pipeline with GPS clustering, temporal consistency checks, and dynamic reputation-based weighting mechanisms.
- Implemented an automated feedback loop for continuous model retraining and a low-latency pub/sub architecture for live bus location, ETA, and occupancy updates.

CuraNex - AI Demand Forecasting & Inventory Optimization Platform | [Project](#)

Technical Stack - Python, React (TypeScript), TensorFlow, PyTorch, XGBoost, LightGBM

- Led development of an AI-driven demand forecasting platform for AITHON 2026 (MVP phase) organized by Hemas Holdings PLC, combining deep sequence models (TFT, NBEATS) and gradient boosting with recursive feature engineering to predict stock requirements and detect inventory anomalies, optimizing supply chain efficiency.
- Designed an interactive analytics dashboard with forecast visualization, anomaly alerts, and inventory recommendations to support decision-making.

Portfolio Website With AI Assistant | [Live Website](#) | [Project](#)

Technical Stack – Next.js 15, Tailwind CSS, Hugging Face, Upstash Vector DB, Groq (LLaMA 3.1), Redis

- Engineered a portfolio website with an AI assistant using Retrieval-Augmented Generation (RAG) to answer user queries based on indexed resume and GitHub data.
- Built a serverless AI pipeline integrating embeddings, vector search, and Redis-based session memory with automatic conversation summarization.
- Developed a responsive and interactive user interface utilizing Next.js, Framer Motion, and GSAP for fluid animations and seamless transitions.

CodeMart – Digital Project Marketplace | [Project](#)

Technical Stack – React (TypeScript), ASP.NET Core, Entity Framework Core, PostgreSQL (Supabase), Docker, Stripe API

- Developed a full-stack digital marketplace for buying and selling software projects with secure authentication, payment integration, and role-based access control.
- Built a RESTful backend supporting project listings, order management, reviews, and transaction workflows.
- Engineered an interactive React frontend using GSAP animations, Three.js and modern UI components to deliver a engaging user experience.

Whisper – Full-Stack Secure Messaging App | [Project](#)

Technical Stack – React (TypeScript), Java, Spring Boot, PostgreSQL, WebSockets, JWT, Tailwind CSS, Cloudinary

- Developed a full-stack secure web chat application for private communication, featuring OTP-based registration and real-time messaging.
- Engineered a robust backend supporting real-time messaging workflows, secure authentication, and integrated Cloudinary for optimized media and file uploads.
- Built a responsive React frontend implementing real-time chat interfaces and contact management UX.

CERTIFICATES

- Deep Learning Specialization | DeepLearning.AI | Credential ID: [PA5O7P3AQ85C](#)
- AI & ML Engineering – Professional Certificate | Microsoft | Credential ID: [DZ6XR637YI5Q](#)
- AWS Cloud Technical Essentials | AWS | Credential ID: [FDLMXF9TS4GQ](#)
- Agentic AI with LangChain and LangGraph | IBM | Credential ID: [AAB65UO5XGDI](#)
- IBM Cloud Essentials | IBM | Credential ID: [ededb5b52cb14d2e97a793e6a9d9ca6f](#)
- AI/ML Engineer – Stage 2 | SLIIT Faculty of Computing | Credential ID: [txqnjcawyu](#)

ACHIEVEMENTS

- **KelaniXtreme – 1st Place (IEEE Student Branch University of Kelaniya)**
- **AITHON 2026 – Top 30 Finalist & MVP Phase (Hemas Holdings PLC)**
- **Research Symposium (FCTSRS 2025) – Selected for Presentation “Project Nexus”**
- **SpiritX Inter University Development Competition – Stage 1 Xcelerate (MoraSpirit)**
- **IX’25 Designathon – Top 20 Team (IEEE Student Branch of IIT)**

EXTRA-CURRICULAR

- Team Member – IEEE EPICS Global, Computerization of Homagama Base Hospital Eye Clinic (2026)
- Member – IEEE Sri Lanka Section (2024 – 2025)
- Member – IEEE Computer Society, University of Kelaniya (2023 – Present)
- Member – Computer Science Student Association (2023 – Present)
- Member – Free and Open-Source Software Community (2024 – Present)
- Rugby – 2nd Runners up (Inter Faculty Tournament 2025)